FPGA Beamformer

Stewart Nash

2026

Chapter 1

Adaptive Filtering

"Filters that adapt to a changing environment are discussed. The stochastic Wiener filtering and deterministic least-squares problems are set up, and the similarity between them is pointed out. First, a block-processing approach is taken in the solution to these two problems. Second, a fading memory (recursive) approach is taken. Then the application of these algorithms to the adaptive beamforming (ABF) problem is considered. The chapter appendix provides background material on conventional beamforming (CBF)."

1.0 Introduction

Numerous applications exist that require a linear filtering operation. Often, the nature of that filtering task is time-varying in some nondeterministic fashion owing to nonstationarity of the underlying time series. In such situations, a filter that can adapt to a changing environment is needed. Examples include adaptive noise canceling, line enhancing, frequency tracking, and channel equalization. Beyond the discussion here, additional material on adaptive filtering can be found in references 1 through 16.

1.1 The Stochastic Wiener Filtering and Deterministic Least Squares Problem

The problem of interest is described in Figure 11.1. The goal is to filter the time series $x[n]$ with a causal, finite impulse response (FIR) digital filter to yield an estimate of the time series $s[n]$. The error in that estimate is denoted $e[n]$

$$e[n] = s[n] - \hat{s}[n] = s[n] + \sum_{k=0}^{N} h_k x[n-k] \tag{1.1}$$

Written in matrix form, Eq. 11.1 becomes

$$e[n] = s[n] + [h_0\, h_1\, \ldots\, h_N] \begin{bmatrix} x[n] \\ x[n-1] \\ \vdots \\ x[n-N] \end{bmatrix} \tag{1.2}$$

or, in vector notation,

$$e[n] = s[n] + \mathbf{h}^T \mathbf{x} \tag{1.3}$$

The solution to this problem for the unknown filter vector $\mathbf{h}$ can be approached in two ways, depending on the optimality criterion chosen

$$\min_{\mathbf{h}} E\left[e^2[n]\right] \tag{1.4}$$

(Wiener filtering problem)

$$\min_{\mathbf{h}} \sum |e[n]|^2 \tag{1.5}$$

(Least squares problem) The Wiener filtering problem takes a stochastic viewpoint where the filter is optimized in an expected value sense (minimum mean-squared error). As an alternative, the least-squares problem views the time series involved as deterministic and the filter is optimized for the specific sequences of record (minimum sum-squared error). Both viewpoints give rise to similar mathematical expressions for $\mathbf{h}$.

The distinction between the Wiener filtering and least squares problems is made in the definitions of $R_{ss}[0]$, $\mathbf{v}$ and $\mathbf{T}$ where

$$\mathbf{v} = \begin{bmatrix} R_{xs}[0] \\ R_{xs}[1] \\ \vdots \\ R_{xs}[N] \end{bmatrix} \tag{1.6}$$

$$\mathbf{T} = \begin{bmatrix} R_{xx}[0] & R_{xx}^*[1] & \cdots & R_{xx}^*[N-1] & R_{xx}^*[N] \\ R_{xx}[1] & R_{xx}[0] & \cdots & R_{xx}^*[N-2] & R_{xx}^*[N-1] \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ R_{xx}[N-1] & R_{xx}[N-2] & \cdots & R_{xx}[0] & R_{xx}^*[1] \\ R_{xx}[N] & R_{xx}[N-1] & \cdots & R_{xx}[1] & R_{xx}[0] \end{bmatrix} \tag{1.7}$$

For the Wiener filtering problem

$$R_{ss}[0] = E[s[n]s^*[n]] \tag{1.8}$$

$$\mathbf{v} = \{E[s[n]x^*[n-j]]\} = \{R_{sx}[j]\} \tag{1.9}$$

$$\mathbf{T} = \{E[x[n-k]x^*[n-j]]\} \tag{1.10}$$

And, for the least squares problem

$$R_{ss}[0] = \sum_n s[n]s^*[n] \tag{1.11}$$

$$\mathbf{v} = \{\sum_n s[n]x^*[n-j]\} = \{v_j\} \tag{1.12}$$

$$\mathbf{T} = \{\sum_n x[n-k]x^*[n-j]\} \tag{1.13}$$

The next step in solving for the unknown filter vector $\mathbf{h}$ is to write an expression for squared error based on Eq. 11.3

$$|e[n]|^2 = s^*[n]s[n] + s^*[n]\mathbf{h}^T\mathbf{x} + \mathbf{h}^\dagger\mathbf{x}^*s[n] + \mathbf{h}^\dagger\mathbf{x}^*\mathbf{h}^T\mathbf{x} \tag{1.14}$$

By making use of the definitions of $R_{ss}[0]$, $\mathbf{v}$, and $\mathbf{T}$ the average squared error $\varepsilon(\mathbf{h})$ can be derived from Eq. 11.14 for both the Wiener filtering and least squares problems as

$$\varepsilon_N(\mathbf{h}) = R_{ss}[0] + \mathbf{v}^\dagger\mathbf{h} + \mathbf{h}^\dagger\mathbf{v} + \mathbf{h}^\dagger\mathbf{T}^T\mathbf{h} \tag{1.15}$$

Now, minimizing $\varepsilon(\mathbf{h})$ with respect to $\mathbf{h}$ leads to the following equation

$$0 = \mathbf{v} + \mathbf{T}^T\mathbf{h} \tag{1.16}$$

or

$$\mathbf{T}^T\mathbf{h} = -\mathbf{v} \tag{1.17}$$

Expressing Eq. 11.17 in expanded form wil be of interest in the next section

$$\begin{bmatrix} R_{xx}[0] & R^*_{xx}[1] & \cdots & R^*_{xx}[N-1] & R^*_{xx}[N] \\ R_{xx}[1] & R_{xx}[0] & \cdots & R^*_{xx}[N-2] & R^*_{xx}[N-1] \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ R_{xx}[N-1] & R_{xx}[N-2] & \cdots & R_{xx}[0] & R^*_{xx}[1] \\ R_{xx}[N] & R_{xx}[N-1] & \cdots & R_{xx}[1] & R_{xx}[0] \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_{N-1} \\ h_N \end{bmatrix} = -\begin{bmatrix} R_{xs}[0] \\ R_{xs}[1] \\ \vdots \\ R_{xs}[N-1] \\ R_{xs}[N] \end{bmatrix} \tag{1.18}$$

or

$$\sum_{j=0}^{N} R_{xx}[j-i]h_j = -R_{xs}[j] \quad i = 0, 1, \ldots, N \tag{1.19}$$

Now, solving Eq. 11.17 for $\mathbf{h}$ yields

$$\mathbf{h} = -(\mathbf{T}^T)^{-1}\mathbf{v} \tag{1.20}$$